# UpCharika Medical AI - System Folder Structure

Complete Technical Architecture, File Directory Catalog & Security Blueprint

## 1. Master Directory Tree Structure

The complete root hierarchy of Medical_chatbot/ organized by functional subsystem:

```
Medical_chatbot/
+-- medibot.py                    [Main Web App & User Interface (Streamlit Framework)]
+-- create_memory_for_llm.py      [Offline Ingestion: 9 Books -> 77,636 Chunks -> FAISS]
+-- style.css                     [Custom Medical Theme: Emerald Glassmorphism & Layout]
+-- requirements.txt              [Exact Python Dependencies & Version Manifest]
+-- users.json                    [User Authentication Store: Usernames & Hashed Passwords]
+-- .env                          [Local Secret Storage: Private GROQ_API_KEY]
+-- .gitignore                    [Security Filter: Blocks .env, vectorstore, models from Git]
+-- README.md                     [GitHub Project Documentation & Deployment Guide]
+-- vectorstore/                  [Knowledge Base Storage Layer]
|   +-- db_faiss/
|   |   +-- index.faiss           [77,636 Vectors (384-D, float32) = 119 MB]
|   |   +-- index.pkl             [Docstore: English Text Chunks, Source Books, Pages = 79 MB]
|   +-- db_faiss.zip              [Compressed DB Archive for Google Drive / Cloud Sync]
+-- .streamlit/
|   +-- config.toml               [Theme Colors, Primary Font, Streamlit Server Port]
|   +-- secrets.toml              [Streamlit Community Cloud Secret Storage (Production)]
+-- models/
|   +-- .cache/huggingface/       [Local Cache: all-MiniLM-L6-v2 Embeddings Weights (~80 MB)]
|   +-- mistral/                  [Historical Offline LLM Experiment: mistral .gguf model]
+-- medbot_env/ (or venv/)        [Isolated Python Virtual Environment (Site Packages)]
+-- __pycache__/                  [Python Bytecode Compilation Speed Cache (.pyc)]
+-- .history/                     [VS Code Local History Automated Undo Snapshots]
+-- .git/                         [Git Version Control Database & Commit History]
```

## 2. Complete Component & Directory Matrix

Detailed functional analysis of every file and folder in the project:

| Item Name | Type & Size | Role & Core Responsibility | Git Status |
|---|---|---|---|
| **medibot.py** | Python Script (~76 KB) | Main application script. Launches Streamlit web UI, multi-model switcher (Qwen, GPT-OSS, Llama-3), FAISS vector search, Groq API calling, emergency SOS cards, and PDF report downloads. | Tracked (Git) |
| **create_memory_for_llm.py** | Python Script (~3.3 KB) | Offline ETL ingestion pipeline. Reads 9 medical PDFs via PyMuPDF, chunks text (1,000 chars, 100 overlap), calculates 384-D vectors via all-MiniLM-L6-v2, and builds FAISS index. | Tracked (Git) |
| **vectorstore/db_faiss/index.faiss** | Binary Index (119 MB) | Facebook AI Similarity Search index containing exactly 77,636 vector embeddings of 384 dimensions each (29.8M float32 values) for sub-2ms nearest-neighbor lookups. | Ignored (Git) |
| **vectorstore/db_faiss/index.pkl** | Pickle Docstore (79 MB) | Serialized docstore holding chunk texts and source citation metadata (PDF book title, specific page number). Converts winning vector IDs into readable medical text. | Ignored (Git) |
| **vectorstore/db_faiss.zip** | Zip Archive (~180 MB) | Compressed package of index.faiss + index.pkl uploaded to Google Drive. Enables Streamlit Cloud to download the knowledge base on deployment. | Ignored (Git) |
| **style.css** | CSS Stylesheet (~20 KB) | Custom UI styling injected into Streamlit. Implements medical glassmorphism, teal/emerald gradients, glowing cards, responsive flex layouts, and styled bot chat bubbles. | Tracked (Git) |
| **.streamlit/config.toml** | TOML Config (<1 KB) | Streamlit web app configuration: defines primary theme color (#0F766E), background colors, and typography settings across client browsers. | Tracked (Git) |

| Item Name | Type & Size | Role & Core Responsibility | Git Status |
|---|---|---|---|
| **.streamlit/secrets.toml** | TOML Secrets (<1 KB) | Cloud secret storage holding GROQ_API_KEY when running on Streamlit Community Cloud. Safely omitted from public repositories. | Ignored (Git) |
| **.env** | Config File (<1 KB) | Local environment variables file storing GROQ_API_KEY for local workstation development. Loaded via python-dotenv; never pushed to remote Git. | Ignored (Git) |
| **users.json** | JSON Store (<1 KB) | Local patient user store containing registration credentials, usernames, and hashed passwords for session authentication. | Tracked (Git) |
| **requirements.txt** | Text Manifest (<1 KB) | List of all 8 core dependencies (streamlit, groq, langchain, langchain-community, langchain-huggingface, sentence-transformers, faiss-cpu, pymupdf). | Tracked (Git) |
| **.gitignore** | Git Exclude (<1 KB) | Security & size exclusion rules. Instructs Git to ignore .env, secrets.toml, vectorstore/, models/, medbot_env/, and __pycache__/. | Tracked (Git) |
| **models/** | Directory (Cached Data) | Holds cached Hugging Face model weights (.cache/huggingface) and older local GGUF Mistral LLM experiments. Runs locally without internet after initial download. | Ignored (Git) |
| **__pycache__/** | Directory (Bytecode) | Python automatic compilation directory containing .pyc bytecode files for instant script startup. | Ignored (Git) |
| **.history/** | Directory (Dev Backups) | VS Code Local History extension checkpoints saved automatically on file modifications for developer undo safety. | Ignored (Git) |

## 3. Security Architecture: Protecting Credentials & Managing Bloat

Modern cloud best practices dictate that sensitive secrets (API keys) and massive binary datasets must never be checked into version control. In UpCharika, this is enforced by .gitignore.

| Category | Excluded Files / Folders | Technical Justification |
|---|---|---|
| **API Credentials** | .env<br>.streamlit/secrets.toml | Prevents leak of private Groq API keys to public repositories. Protects API quota and user confidentiality. |
| **Vector Store DB** | vectorstore/db_faiss/ (119 MB + 79 MB) | GitHub hard-rejects any file > 100 MB. The vector store is hosted on Google Drive (vs.zip) and fetched via cloud script. |
| **Virtual Envs** | medbot_env/, venv/ | Machine-specific binary directories (> 1 GB). Portable deployment is handled cleanly via requirements.txt. |
| **AI Models Cache** | models/, *.gguf | Weights are automatically downloaded and cached by Hugging Face on the target machine; no need to track in Git. |

## 4. Essential Git Commands for File Security

Commands used to audit, enforce, and untrack files in Git:

```
# 1. Verify if .env is actively ignored by .gitignore:
git check-ignore -v .env
# Output: .gitignore:11:.env    .env

# 2. Check current working directory status:
git status
# (.env, vectorstore, medbot_env will NOT appear in untracked files list)

# 3. Untrack a file accidentally committed without deleting local file:
git rm --cached .env
git commit -m "Remove .env from git tracking"
git push origin main
```

## 5. Knowledge Base: The 9 Indexed Medical Textbooks

The 77,636 chunks inside index.faiss and index.pkl were extracted from these 9 authoritative medical sources:

```
1. Davidson's Principles & Practice of Medicine (Internal Medicine & Diagnostics)
2. Harrison's Principles of Internal Medicine, 21st Edition (Vols 1 & 2 - Gold Standard)
3. Joint Formulary Committee - British National Formulary (BNF 76, 2018)
4. KD Tripathi - Essentials of Medical Pharmacology (Drug Classifications & Dosages)
5. Medical Book - The Gale Encyclopedia of Medicine (Patient-Facing Symptoms & Causes)
6. Oxford Handbook of Clinical Medicine (10th Edition - Acute Emergencies)
7. Rang & Dale's Pharmacology (Elsevier 2019 - Mechanisms of Action)
8. Robbins and Cotran Pathologic Basis of Disease (Disease Mechanisms & Etiology)
9. Standard Treatment Guidelines (Government Clinical Protocols & Management)
```

## 6. Viva & Oral Defense Q&A;: System Structure

### Q1: Why do you have two Python files (medibot.py and create_memory_for_llm.py)?

create_memory_for_llm.py is an offline, one-time ETL script that reads the 9 heavy medical books and generates the mathematical vector index. medibot.py is the online production web application that runs daily, loading the pre-built FAISS index in 20 milliseconds to answer patient questions.

### Q2: Why is index.faiss exactly 119 MB?

It stores 77,636 text chunks embedded into 384 dimensions using all-MiniLM-L6-v2. Math: 77,636 chunks x 384 dimensions x 4 bytes (float32) = 119,252,736 bytes = 119 MB.

### Q3: What is the difference between index.faiss and index.pkl?

index.faiss stores only the mathematical vectors for distance calculation (no text). index.pkl is a serialized Python docstore containing the original English paragraphs, book titles, and page numbers so the bot can cite its sources.

### Q4: Why is .env kept in .gitignore while requirements.txt is committed?

.env contains private API credentials (GROQ_API_KEY) which must never be exposed publicly. requirements.txt contains no secrets, only software package names, allowing any developer to replicate the environment safely.

### Q5: Can __pycache__ and .history be safely deleted?

Yes, 100%. __pycache__ contains auto-generated Python bytecode (.pyc) that Python regenerates automatically. .history contains local editing undo checkpoints created by the VS Code extension.